

MUTUAL FUND ANALYTICS AND DASHBOARD

Bluestock Fintech Capstone Project

REPORT

Introduction

Mutual funds are one of the most popular investment options available today. Investors rely on various factors such as returns, risk, portfolio composition, and market trends before making investment decisions. The objective of this project is to analyze mutual fund data using Python and Power BI and provide insights into fund performance, investor behavior, and industry trends. The project follows a complete workflow starting from data cleaning and ending with dashboard creation and reporting.

Problem Statement

Large amounts of mutual fund data are generated every day, making manual analysis difficult and time-consuming. Different datasets contain information related to NAV history, AUM, investor transactions, SIP inflows, and portfolio holdings. The objective of this project is to automate data processing and develop an analytical system that can provide meaningful insights and visualizations for better understanding of mutual fund performance and market trends.

Objectives

The main objectives of this project are:

- To build an ETL pipeline for extracting, transforming, and loading mutual fund datasets.
- To clean and validate raw data for ensuring consistency and reliability.
- To design a SQLite database for storing processed datasets.
- To perform exploratory data analysis and identify important patterns and trends.

- To evaluate fund performance using financial metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, and Beta.
- To analyze investment risk using measures such as Value at Risk and Maximum Drawdown.
- To study investor behavior and SIP continuity patterns.
- To create an interactive Power BI dashboard for visualizing important insights.
- To generate reports and recommendations based on analytical findings.

Data Sources

The project utilizes ten datasets containing information related to various aspects of mutual funds and investor activities.

The datasets used are:

1. Fund Master Data
2. NAV History
3. AUM by Fund House
4. Monthly SIP Inflows
5. Category Inflows
6. Industry Folio Count
7. Scheme Performance
8. Investor Transactions
9. Portfolio Holdings
10. Benchmark Indices

These datasets cover information from 2022 to 2026 and provide details regarding fund characteristics, performance metrics, investor transactions, market benchmarks, and portfolio composition. Together, they provide a comprehensive view of the mutual fund industry and form the foundation for analytical and visualization tasks performed in this project.

Tools used

Programming Language	- Python
Data Processing	- Pandas, NumPy
Visualization	- Matplotlib, Seaborn, Plotly
Database	- SQLite
Notebook Environment	- VScode
Dashboard Tool	- Power BI
Version Control	- Git and GitHub
Development Environment	- VScode

Methodology

The project followed a systematic approach to transform raw mutual fund datasets into meaningful insights and visualizations.

Data Collection

Multiple datasets related to mutual fund schemes, NAV history, investor transactions, SIP inflows, portfolio holdings, and benchmark indices were collected in CSV format. These datasets provide information about different aspects of the mutual fund industry.

Data Cleaning and Validation

The collected datasets contained missing values, inconsistent formats, and duplicate records. Python and Pandas were used to standardize date formats, remove duplicate entries, validate numerical fields, and ensure consistency across datasets.

Database Design

A star schema was designed and implemented using SQLite. Dimension tables and fact tables were created to store information efficiently and support analytical queries.

Exploratory Data Analysis

Different visualizations were generated to understand trends in NAV movements, AUM growth, SIP inflows, investor demographics, and category-wise fund performance.

Performance and Risk Analysis

Financial metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Value at Risk were calculated to evaluate fund performance and risk characteristics.

Dashboard Development

Power BI was used to build interactive dashboards that provide insights into industry trends, fund performance, investor behavior, and market movements.

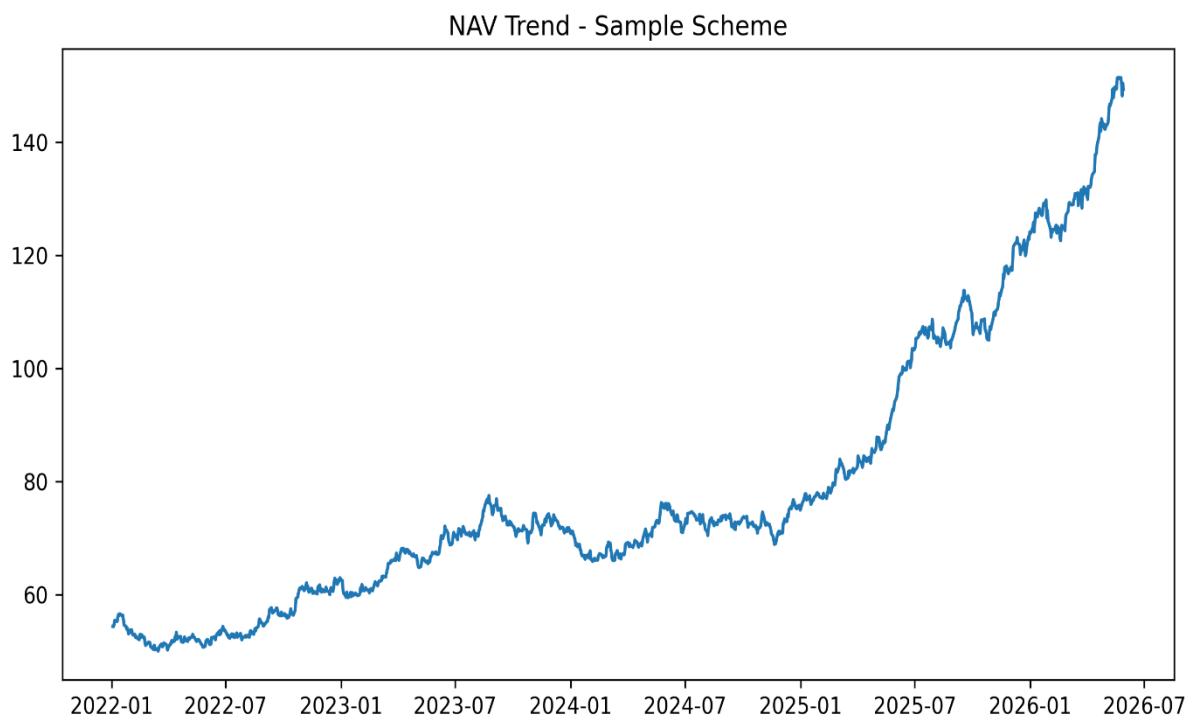
Database Design

A SQLite database was created to store processed datasets in a structured manner. A star schema approach was used to improve query performance and maintain relationships between tables. The database consists of dimension tables containing descriptive information and fact tables containing transactional and analytical data. Primary keys and foreign keys were defined to maintain data integrity and consistency. The database design helped in performing analytical queries efficiently and served as the foundation for dashboard development.

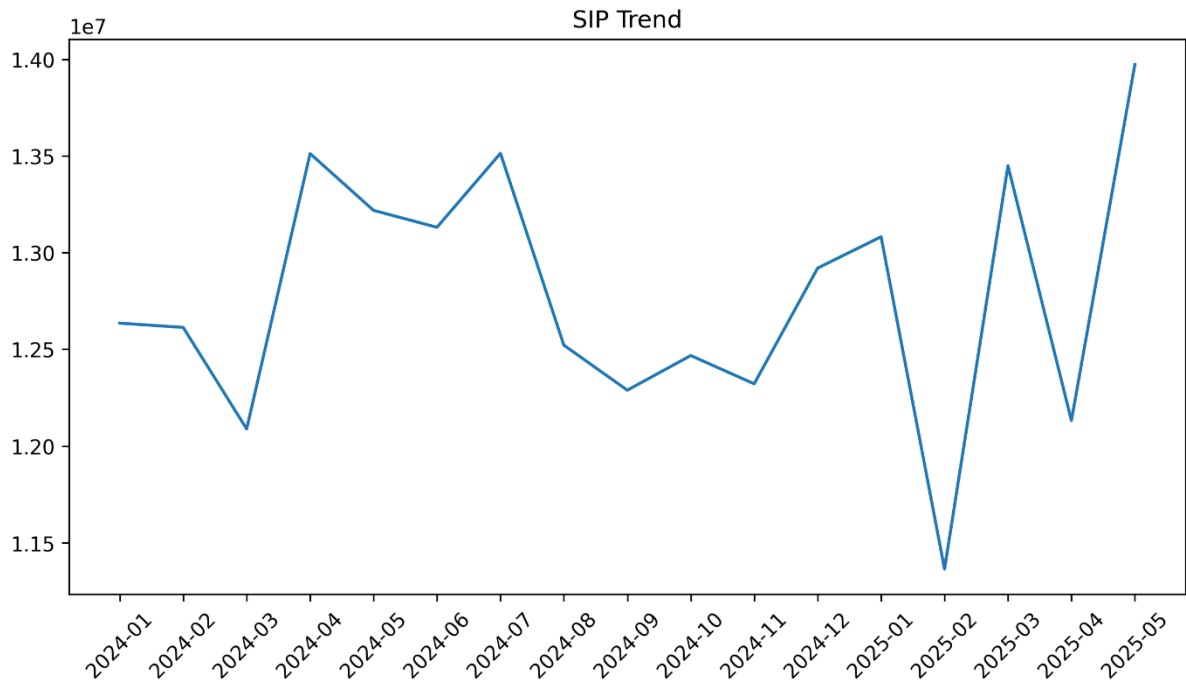
Exploratory Data Analysis

Exploratory Data Analysis was performed to understand trends and patterns present in the mutual fund datasets. Various charts and visualizations were generated to analyze fund performance and investor behavior.

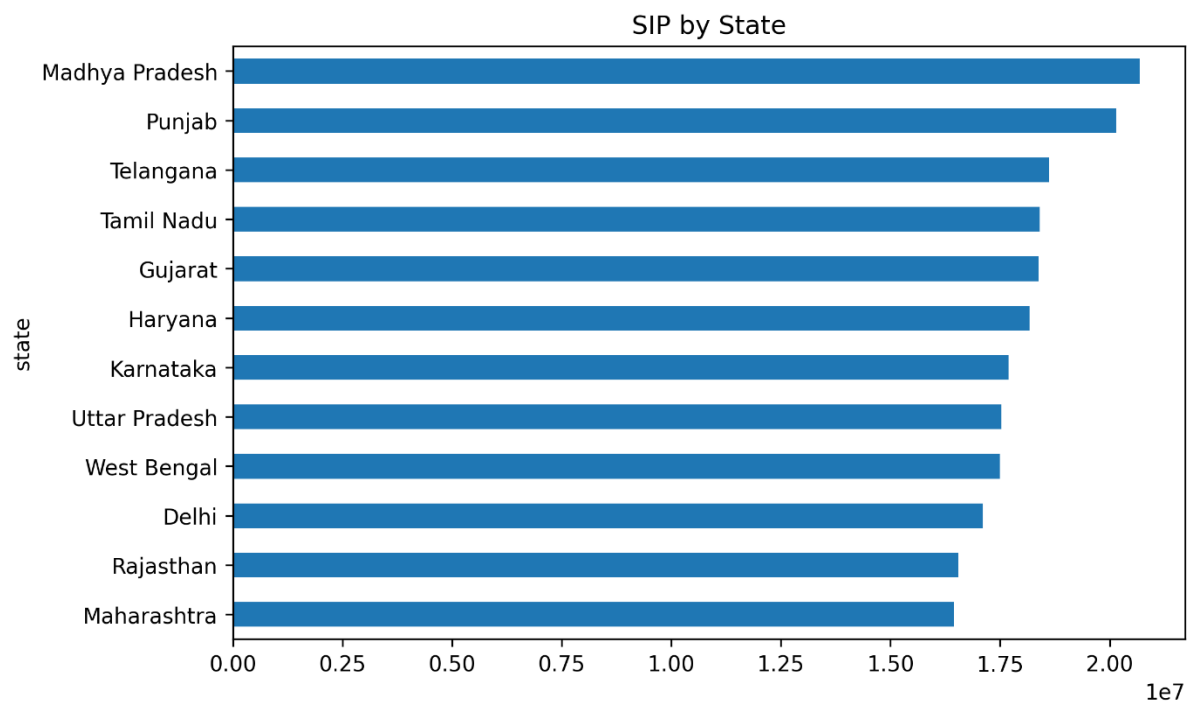
NAV trends were studied to observe changes in scheme values over time. AUM growth was analyzed to understand the expansion of fund houses. SIP inflow trends and category-wise investments were examined to identify investor preferences. Demographic analysis was also performed to study investment patterns across different age groups and geographical regions.



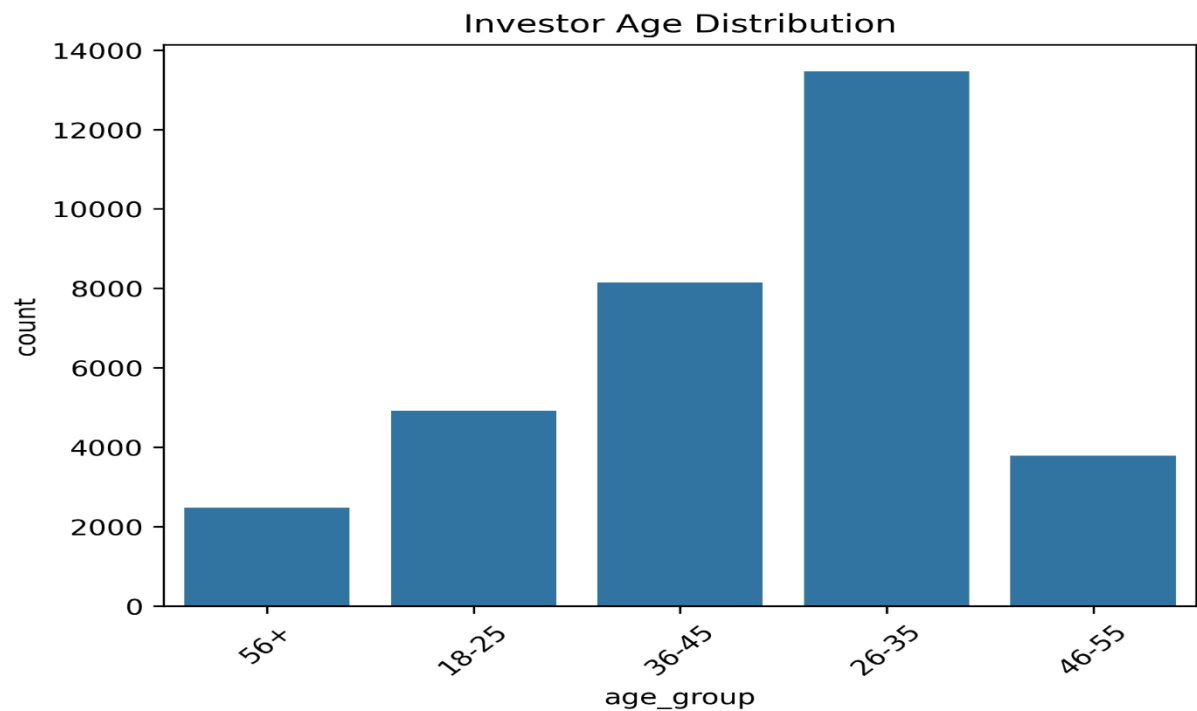
The NAV trend chart shows the movement of scheme values over time and highlights overall growth patterns.



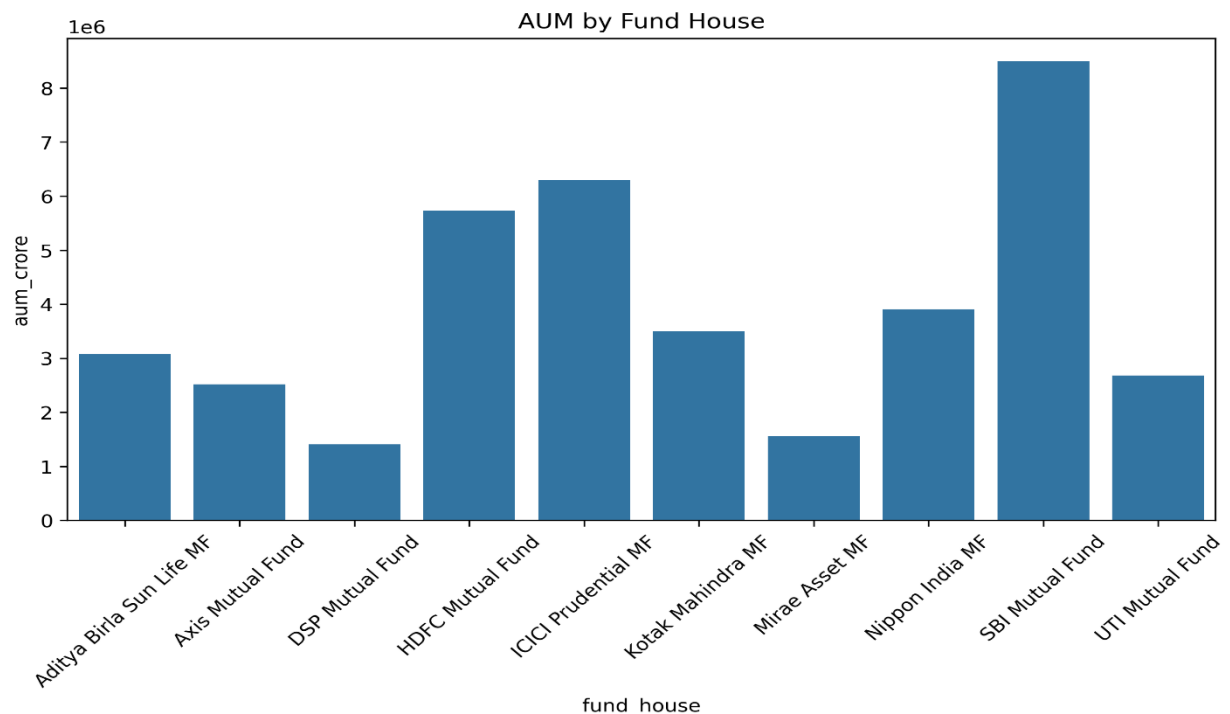
SIP inflows showed steady growth during the study period, indicating increasing participation of investors.



Certain states contributed significantly higher transaction volumes compared to others.



Investment participation varied across age groups, with middle-aged investors contributing the largest share.



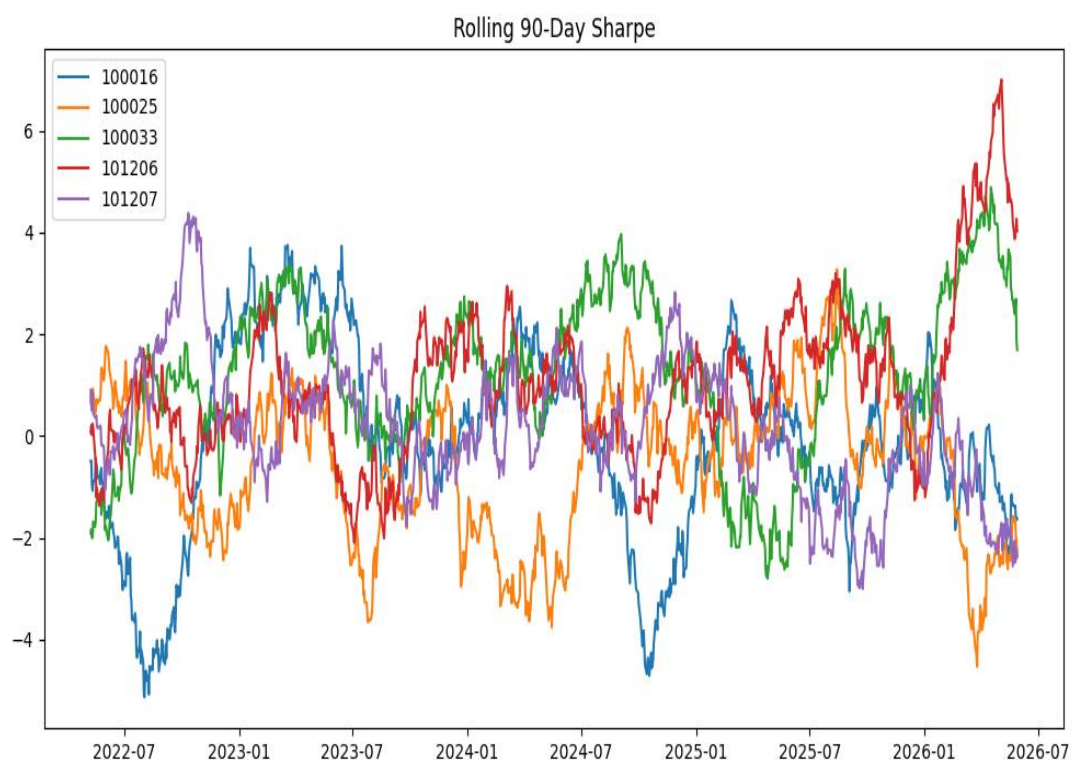
The AUM growth chart highlights the expansion of major fund houses over the years. Larger fund houses maintained a dominant market share, reflecting strong investor confidence and increasing asset accumulation.

Performance Analytics

Performance analysis was carried out using several financial metrics to evaluate mutual fund schemes. CAGR was used to measure annualized returns, while Sharpe Ratio and Sortino Ratio were used to analyze risk-adjusted returns.

Alpha and Beta values helped compare fund performance with benchmark indices. Maximum Drawdown was calculated to understand the maximum decline experienced by funds during adverse market conditions.

These metrics provided a comprehensive understanding of return and risk characteristics across different schemes.



Rolling Sharpe Ratio helps evaluate how risk-adjusted returns changed over time for selected funds.

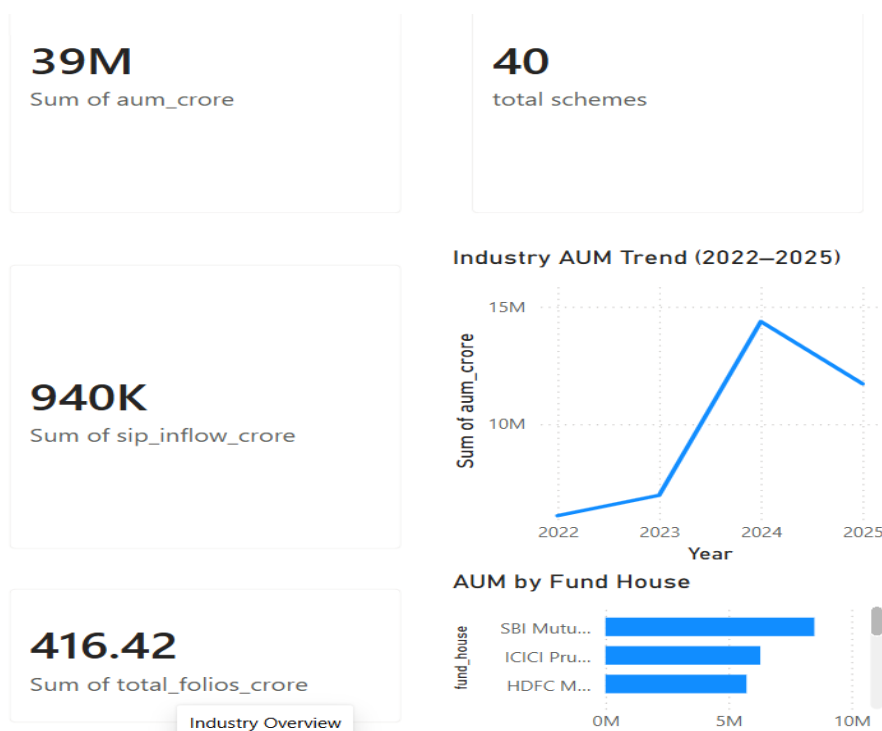
Advanced Risk Analysis

Risk analysis was performed to measure downside risk and portfolio concentration. Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR) were calculated using daily returns. Rolling Sharpe Ratio was used to study changes in risk-adjusted performance. Investor cohort analysis and SIP continuity analysis helped understand investor behavior and identify irregular investment patterns. Sector concentration was measured using the Herfindahl-Hirschman Index (HHI), which provided insights into portfolio diversification.

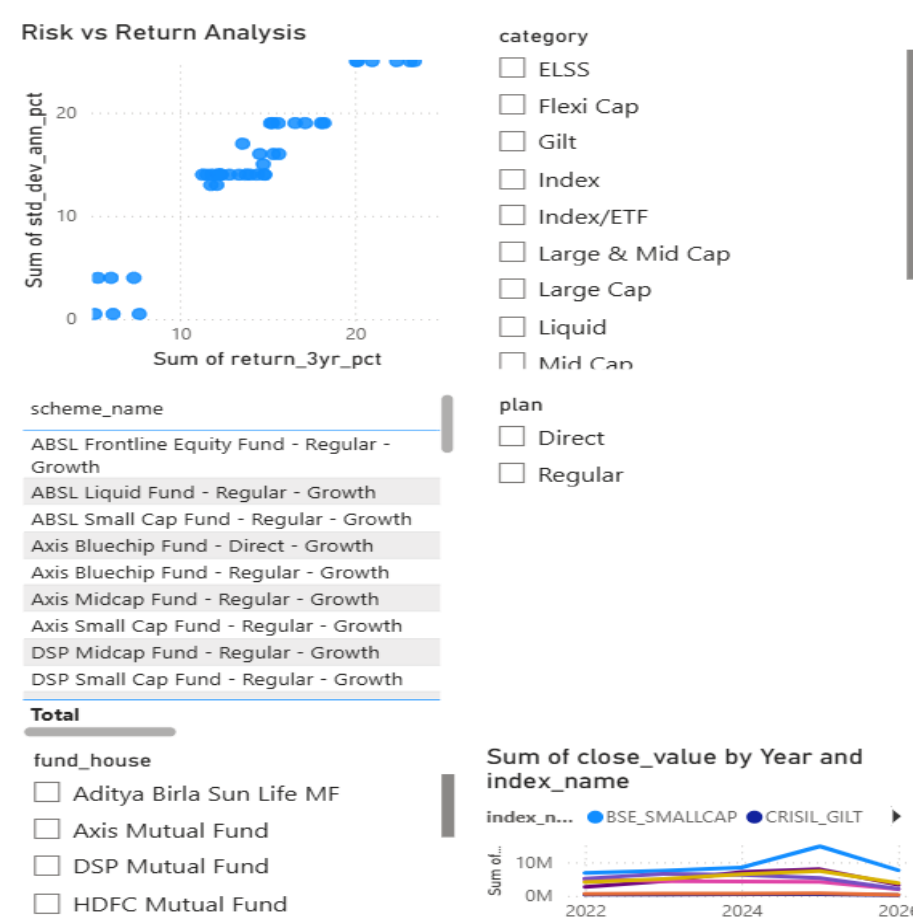
Power BI Dashboard

An interactive Power BI dashboard was developed to visualize various aspects of the mutual fund industry. Four dashboard pages were created to provide insights into industry performance, fund analysis, investor behavior, and SIP trends. The dashboard includes KPI cards, trend charts, scatter plots, demographic analysis, benchmark comparisons, and interactive slicers. These visualizations enable users to analyze data efficiently and gain a better understanding of market trends.

Page 1 – Industry Overview



Page 2 – Fund Performance



plan

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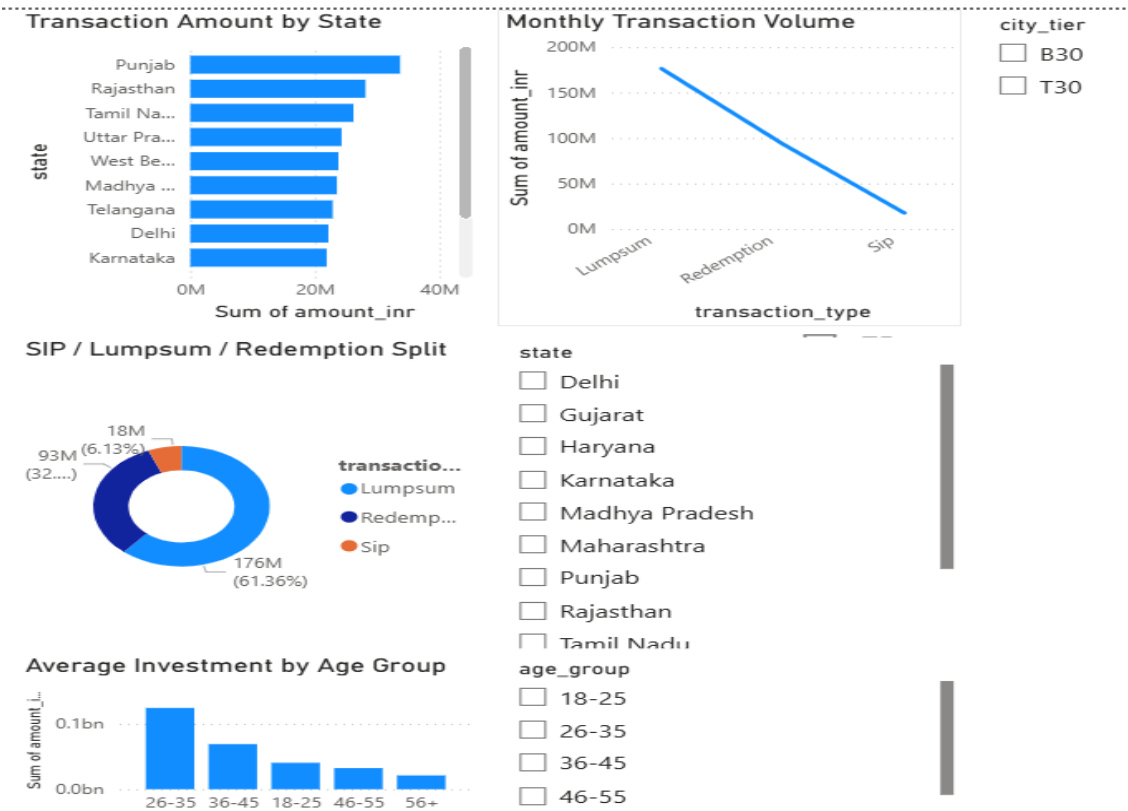
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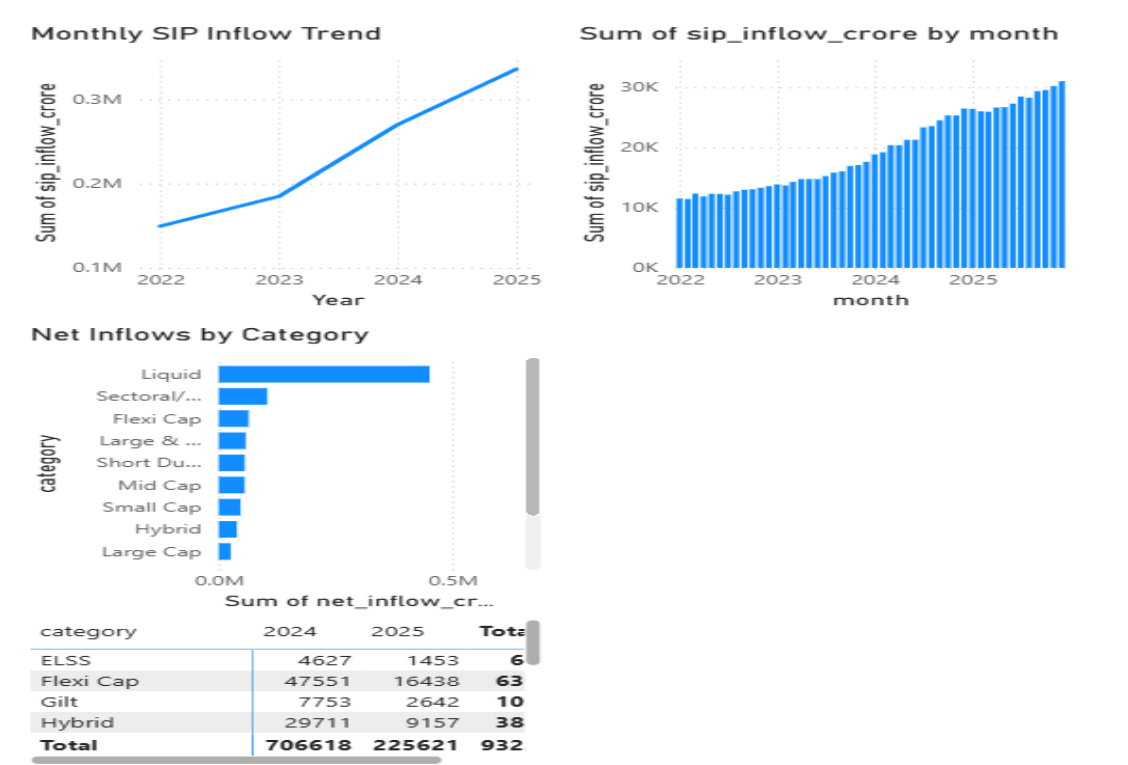
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Page 3 – Investor Analytics



Page 4 – SIP and Market Trends



Key Findings

- Mutual fund industry AUM showed consistent growth.
- SIP inflows increased steadily during the study period.
- Some funds demonstrated superior risk-adjusted returns.
- Recent investor cohorts contributed higher investment amounts.
- Diversified portfolios exhibited lower concentration risk.

Recommendations

- Encourage long-term SIP investments.
- Focus on diversified portfolios.
- Use risk-adjusted metrics while selecting funds.
- Continuously monitor market risk and fund performance.

Conclusion

This project successfully implemented an end-to-end mutual fund analytics framework using Python, SQLite, Jupyter Notebook, and Power BI. Data engineering techniques were used to process and organize raw datasets, while analytical methods helped evaluate fund performance and risk characteristics.

Interactive dashboards and visualizations provided meaningful insights into industry trends, investor behavior, and market movements. Overall, the project demonstrates how data analytics and business intelligence techniques can be effectively applied in the financial domain to support better understanding and decision-making.